

A Hybrid Multi-Stage Deep Learning Model for River Water Quality Prediction and Pollution Anomaly Detection

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Abstract—Water quality time series are nonlinear, nonstationary and irregularly sampled, which limits conventional forecasting methods. This paper presents a hybrid framework coupling Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) with a convolutional–recurrent network and a self-attention layer. Nine measured parameters are each decomposed into intrinsic mode functions, giving an 81-channel input passed through a 1-D CNN, an LSTM and attention pooling to jointly predict all nine. A per-parameter residual baseline additionally drives a 3σ anomaly detector. On 1,085 observations spanning 2016–2024, the model reduces mean absolute error against a persistence baseline on seven of nine parameters, by up to 33%. The point accuracy initially targeted was not attained, with coefficients of determination remaining near or below zero. Decomposing this result, however, shows it stems from amplitude damping rather than from a failure to learn temporal structure. For the seasonally driven parameters the prediction tracks the observed series in correct phase (Pearson $r = 0.56$ for temperature, 0.48 for dissolved oxygen) while reproducing only 65–73% of its variance. The temporal stream recovers *when* parameters move but not *how far*. We argue the missing amplitude information is largely spatial, and that the natural next step is to pair this temporal expert with a spatial expert over remotely sensed catchment features under a learned fusion gate.

Index Terms—Water quality prediction, CEEMDAN, CNN, LSTM, self-attention, anomaly detection, sensor fusion

I. INTRODUCTION

A. Background and Motivation

Water quality refers to the chemical, physical and biological characteristics that determine a water body’s suitability for drinking, irrigation, industry or aquatic life. It directly influences biodiversity, agricultural productivity and public health, yet industrialisation and urbanisation have made contamination by nutrients, heavy metals and organic pollutants a pressing global concern. Accurate prediction of water quality parameters is therefore central to early pollution warning and sustainable resource planning.

Routine monitoring centres on a small set of indicators. *Dissolved oxygen* governs the survival of aquatic organisms, with depressed levels producing fish kills; *pH* controls metal solubility and biological function; *turbidity* measures suspended particulate load and limits light penetration; and *conductivity* and *total dissolved solids* track the ionic content that indicates industrial or saline intrusion. Nutrient species – ammonia,

nitrate and phosphates – drive eutrophication. This work predicts all nine such parameters jointly.

Prediction is difficult for reasons intrinsic to the data. Measurements are *nonlinear*, since parameters interact through coupled biochemical processes: a rise in temperature depresses dissolved oxygen saturation, while nutrient loading drives algal activity that perturbs both. They are *nonstationary*, as seasonal cycles and land-use change shift the underlying distribution. They are *noisy*, and *sparsely sampled*, since routine monitoring reports at multi-day rather than continuous intervals. The events of greatest practical interest, pollution spikes, are also rare and transient.

Signal decomposition offers a principled response to nonstationarity: separating a series into narrow-band oscillatory components makes each closer to stationary and easier to model than the composite. CEEMDAN suits this role, being fully data-driven and requiring no basis selection unlike wavelet methods.

The problem addressed is therefore: given a multivariate record of river water quality sampled at irregular multi-day intervals, predict the next observation for all parameters simultaneously, and flag readings whose deviation from prediction indicates a pollution event rather than ordinary model error. Our objectives are to develop the architecture, quantify its accuracy against interpretable baselines, and derive a grounded anomaly rule from the residual distribution.

B. Related Work

Zhu et al. [1] address nonstationarity by applying CEEMDAN to decompose raw series into intrinsic mode functions, then feeding an LSTM–CNN architecture with self-attention, reporting high accuracy for dissolved oxygen and pH; this directly motivates the architecture adopted here. Yan and Wang [2] model Yellow River Basin stations as graph nodes, combining graph convolution with reinforcement learning to exploit inter-station structure, and outperform LSTM and GCN baselines (RSE = 0.071, CORR = 0.928 for chlorine), though stability degrades during extreme surges.

Beyond station time series, remote sensing supports broader spatial assessment. Shu et al. [3] upscale low-resolution satellite imagery to 0.5 m granularity with a super-resolution CNN before groundwater quality mapping, combining the

enhanced imagery with field measurements of pH, hardness and total dissolved solids across 33 sites. A fuzzy analytic hierarchy process assigns each parameter a weight, yielding a composite groundwater quality index that revealed a negative correlation with groundwater depth. Reconstruction reached 32.4 dB PSNR and 0.91 SSIM.

Sudriani et al. [4] surveyed over 300 papers spanning 2011–2022 to produce the Framework for Integrated Monitoring Systems, a seven-stage pipeline running from data gathering and real-time integration through modelling and early warning to decision support and sensor-placement optimisation. They compare floating buoys, unmanned surface vehicles, fixed stations and remote sensing, and note that although IoT deployment enables high-frequency sensing, ground-based monitoring remains essential for calibration and validation. Persistent gaps they identify – limited sample sizes, incomplete spatial-temporal coverage and a shortage of field validation – apply directly to the present work. Broader reviews [9] establish that catchment-scale spectral information carries signal about nutrient and sediment loading that station histories do not, a point we return to in Section IV.

For classification rather than regression, Pandya et al. [5] stacked decision trees, random forest, XGBoost and neural classifiers under a logistic-regression meta-learner, validated by 5-fold stratified cross-validation across 46 monitoring stations in Gujarat and reaching 96.7% accuracy on a nine-parameter potability task. Their inference cost of under 3 ms per sample makes IoT deployment practical, and a transformer benchmark showed modern architectures transfer to structured tabular data.

Rahu et al. [6] paired an IoT sensor network measuring temperature, pH, turbidity and total dissolved solids with cloud analytics, comparing LSTM, SVR, MLP and NARNet for regression against SVM, XGBoost, decision trees and random forest for classification on 6,000 Rohri Canal samples collected over four months. MLP led on R^2 and random forest exceeded 0.91 classification accuracy. Their observation that carefully curated small datasets can outperform larger noisier ones is relevant here, since our record is small but complete.

C. Research Gaps and Our Response

Four drawbacks recur across this literature, and each shapes a design decision here.

Narrow dataset scope. Several studies train on records too short to contain the seasonal behaviour they predict: 325 samples in Pandya et al. [5], a single month in Yan and Wang [2], which cannot express an annual cycle at all. We evaluate on a nine-year record (2016–2024, 1,085 observations), so multiple full seasonal cycles fall in both the training and test partitions.

Absent baselines. Accuracy is often reported without a trivial reference, so it is impossible to tell whether a complex model beats repeating yesterday’s reading. We report every result against persistence and train-mean baselines, and Section IV shows this substantially changes the interpretation of our own results.

Aggregate metrics that hide failure modes. A single R^2 or RMSE conflates whether a model knows *when* a parameter changes with *how far*, so correct timing with compressed amplitude is indistinguishable from no skill. We report scale-invariant trend correlation alongside an amplitude ratio, which is what lets us diagnose what the model actually learned.

Detection folded into prediction. Expecting one network to forecast both routine variation and rare spikes is unrealistic, since a loss optimised for average-case error is dominated by the former. We separate the tasks, using the predictor only to establish a residual distribution and building a 3σ detector on it.

Addressing these does not by itself guarantee a more accurate model, and Section IV reports that our accuracy target was not met. What it does guarantee is that the resulting evaluation is interpretable: when performance falls short, the reporting is sufficient to identify why.

II. SYSTEM ARCHITECTURE

Effective river monitoring requires a framework that processes noisy sensor data, predicts water quality, detects deviations and exposes results to operators. The proposed system is organised as four layers, each transforming its input into a form the next can consume:

- **Data acquisition and preprocessing** – ingests raw multivariate readings and applies CEEMDAN decomposition, emitting a set of intrinsic mode functions and a residual per parameter.
- **Hybrid prediction model** – consumes the decomposed representation and produces a joint forecast over all nine parameters via a CNN, an LSTM and a self-attention layer.
- **Anomaly detection** – compares observations against predictions and flags deviations exceeding a residual-derived threshold.
- **Service layer** – exposes ingestion and retraining endpoints and schedules periodic retraining to track distributional drift.

The layers are described in turn below. Fig. 1 presents the complete architecture, showing the progression from raw inputs through decomposition and prediction to anomaly attribution and the conversational user interface.

A. Data Acquisition and Preprocessing

This layer ingests raw readings and applies CEEMDAN, which separates a nonstationary signal into intrinsic mode functions (IMFs) and a residual, each IMF capturing an oscillatory mode from high-frequency noise to low-frequency trend:

$$X_{\text{param}}(t) = \sum_{k=1}^N \text{IMF}_k(t) + r_N(t) \quad (1)$$

CEEMDAN operates iteratively: white noise realisations are added to the current residual, the first mode of each realisation is extracted by EMD, and their ensemble mean is taken

WATER QUALITY ANALYSIS & PREDICTION SYSTEM ARCHITECTURE

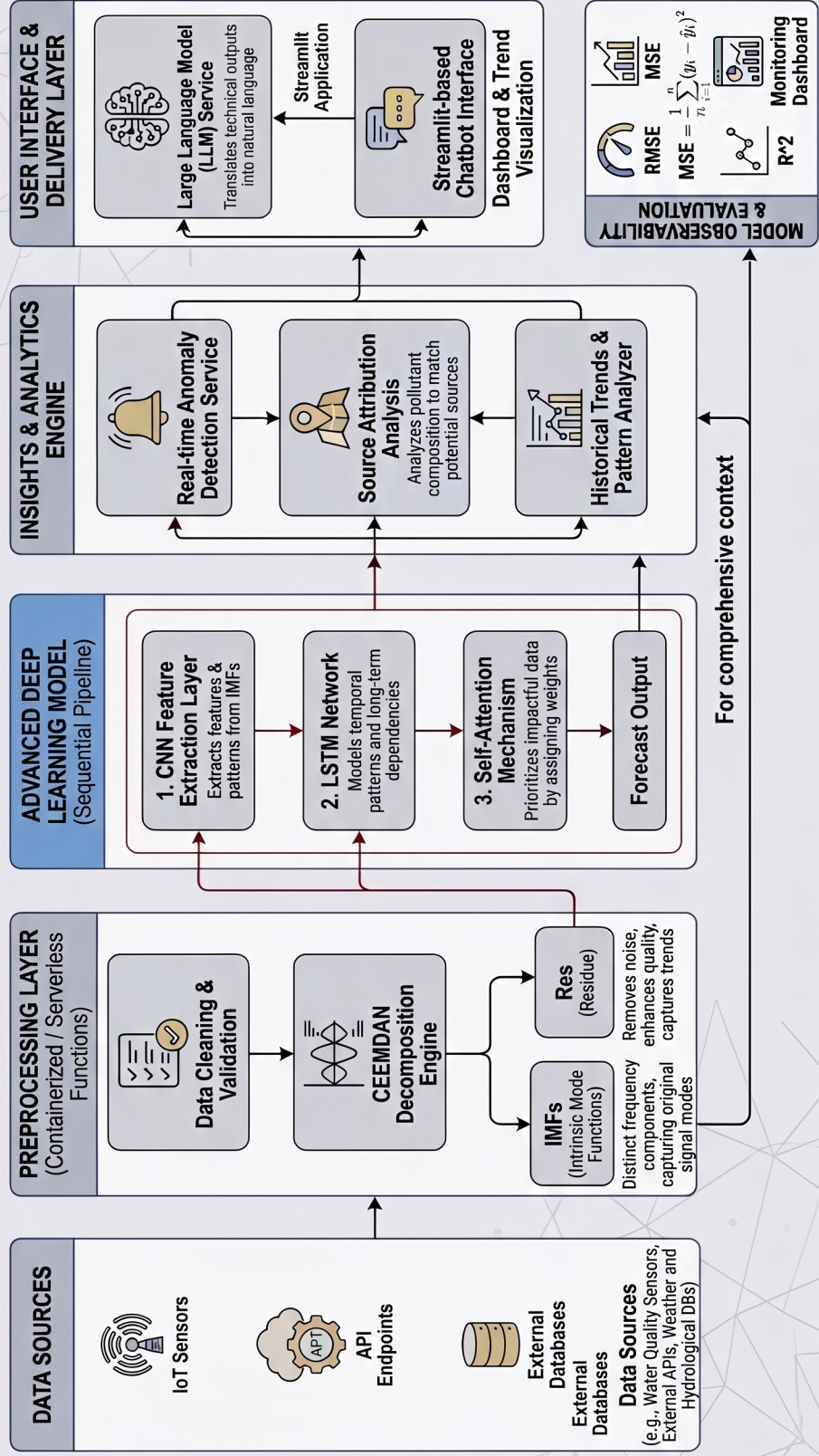


Fig. 1. Water quality analysis and prediction system architecture. Readings enter from sensors, APIs and external hydrological databases; the preprocessing layer cleans them and applies CEEMDAN, emitting intrinsic mode functions and a residual. The deep learning pipeline applies CNN feature extraction, an LSTM for temporal dependency and self-attention for weighting, producing the forecast. The insights engine performs anomaly detection, source attribution and trend analysis, feeding an LLM service and Streamlit chatbot in the delivery layer. Preprocessed signals are also routed directly to the analytics engine for context, and evaluation metrics feed a monitoring dashboard. The diagram describes the target deployment; the evaluation reported in Section IV uses an archival dataset rather than live sensor ingest.

Algorithm 1: CEEMDAN decomposition

Input: signal $x(t)$; trials $M = 100$; noise amplitude ϵ
Output: IMF set $\{c_k\}$ and residual r_N
 $r_0 \leftarrow x$; $k \leftarrow 0$
while *stopping criteria not met* **do**
 $k \leftarrow k + 1$
 for $m \leftarrow 1$ **to** M **do**
 draw white noise w_m
 $d_m \leftarrow \text{EMD}_1(r_{k-1} + \epsilon w_m)$ // first mode
 end
 $c_k \leftarrow \frac{1}{M} \sum_m d_m$ // ensemble mean
 $r_k \leftarrow r_{k-1} - c_k$
end
return $\{c_k\}, r_k$

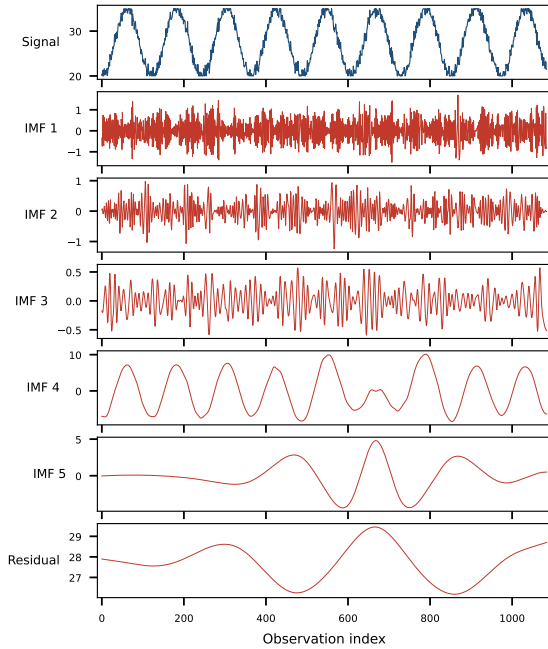


Fig. 2. CEEMDAN decomposition of the temperature series. The first modes carry high-frequency measurement noise; later modes isolate the annual cycle, and the residual captures the long-term trend. Eight modes were extracted for this parameter.

as the next IMF before being subtracted from the residual. Averaging over realisations suppresses the mode-mixing that affects plain EMD, while the adaptive noise schedule keeps the decomposition complete. Algorithm 1 states the procedure. Our implementation uses the EMD-signal package with its default configuration: 100 noise realisations per mode, noise amplitude $\epsilon = 0.005$, and termination when the residual retains less than 5% of the original power or spans a range below 1% of the signal.

Fig. 2 shows the decomposition applied to the temperature series. The leading modes isolate high-frequency variation attributable to measurement noise, intermediate modes recover the annual cycle, and the residual retains the slow trend. This

TABLE I
LAYER FUNCTIONS, TENSOR SHAPES AND PARAMETER COUNTS

Layer	Function	Output	Params
Input	Padded IMF tensor	30×81	0
Conv1D	Local pattern extraction	28×64	15,616
MaxPooling1D	Downsample, suppress jitter	14×64	0
LSTM	Temporal dependency	14×50	23,000
Self-attention	Weighted pooling	50	2,550
Dense	Joint regression	9	459
Total trainable parameters			41,625

separation is what makes each component easier to model than the composite signal.

Because CEEMDAN is adaptive, different parameters yield different IMF counts. Temperature produced eight modes and the maximum across all nine parameters was nine. To obtain a fixed-width tensor the decomposition is zero-padded to the maximum observed, $N = 9$ for this dataset, so that each of the nine parameters contributes nine channels and the network sees a constant 81-channel input.

B. Hybrid Prediction Model

The padded tensor passes through a stacked CNN–LSTM–attention network. The 1-D convolution extracts local morphological features; max pooling reduces sequence length and suppresses jitter; the LSTM models longer-range dependency; and self-attention [7] performs learned weighted pooling over the LSTM output. Given hidden states $\mathbf{x}$:

$$\mathbf{e} = \tanh(\mathbf{x}\mathbf{W} + \mathbf{b}), \quad \alpha = \text{softmax}(\mathbf{e}), \quad \mathbf{z} = \sum_t \mathbf{x}_t \odot \alpha_t \quad (2)$$

so each historical step’s contribution is explicitly weighted, providing interpretability absent from fixed pooling. Unlike scaled dot-product attention, this formulation scores each time step against a single learned projection rather than against every other step, which suits the short 30-step window and keeps the parameter count low.

Table I traces the tensor through the network, with shapes and parameter counts read from the saved model. The convolution dominates the parameter budget because it must span all 81 input channels; the whole network holds 41,625 trainable weights, small enough to train on 844 sequences without severe overfitting, though the loss curve still requires early stopping.

C. Anomaly Detection and Attribution Layer

Rather than predicting anomalies directly, the system monitors residuals. During training the residual standard deviation σ_f is recorded per parameter f , and a reading is flagged when $y_{\text{actual}} - y_{\text{pred}} > 3\sigma_f$. The test is deliberately one-sided, flagging only anomalously high values and only for the six pollution-indicative parameters, since a reading below prediction does not indicate contamination.

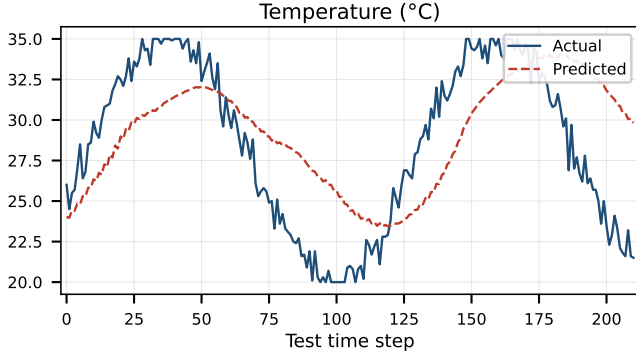


Fig. 3. Predicted versus actual temperature over the 211-step test horizon. The annual cycle is tracked in correct phase, but peaks and troughs are systematically under-shot.

Once an event is flagged, an attribution sub-module examines which parameters moved together. Different industrial processes leave different signatures: a joint rise in conductivity and total dissolved solids suggests saline or chemical effluent, whereas elevated turbidity with nutrient species points toward agricultural runoff or sediment mobilisation. Matching the observed signature against a database of industrial discharge profiles, filtered by geospatial proximity to the affected station, narrows the set of plausible sources. Because proximity and compatibility are grounds for inclusion in a shortlist rather than evidence of discharge, the output is presented as candidate sources for investigation rather than as a determination.

D. Service and User Application Layers

A REST service exposes ingestion and retraining endpoints, with a scheduler triggering quarterly retraining so the model tracks distributional drift as new readings accumulate.

Above this sits the user application layer, which converts model output into a form domain experts and the public can act on. A large language model interprets predictions, alerts and attribution results, and answers natural language questions against them, so that a query such as “*river health on 15 March 2024?*” returns a categorical assessment with its supporting evidence rather than a vector of nine numbers. When an anomaly is active the same channel raises the alert together with the shortlisted sources. A Streamlit front-end hosts this conversation alongside trend visualisations, letting users select a date range, trigger inference and toggle between raw trends, flagged anomalies and the underlying decomposition.

This layer is what makes the residual-based detector usable in practice. A 3σ exceedance on ammonia is not self-explanatory to a non-specialist, whereas a statement that ammonia is anomalously high, that the shortlist contains two facilities upstream, and that the reading is outside the range the model was validated on, is directly actionable.

III. IMPLEMENTATION

Implementation used Python 3 with TensorFlow and Keras, the EMD-signal package for CEEMDAN, NumPy and Pandas,

TABLE II
DATASET SUMMARY (1,085 OBSERVATIONS, 2016–2024)

Parameter	Mean	Std	Min	Max
Temperature (°C)	27.58	5.27	20.00	35.0
pH	7.50	0.15	7.10	8.0
Dissolved oxygen (mg/L)	8.99	1.39	7.00	11.0
Conductivity ($\mu\text{S}/\text{cm}$)	448.22	51.83	293.10	607.4
Turbidity (NTU)	9.85	17.84	2.00	149.8
TDS (mg/L)	290.88	34.90	189.90	407.1
Ammonia (mg/L)	0.149	0.092	0.05	0.5
Nitrates (mg/L)	1.482	0.891	0.50	4.0
Phosphates (mg/L)	0.069	0.048	0.02	0.3

TABLE III
PREPROCESSING PIPELINE

Step	Purpose	Method
Type coercion	Numeric dtype	<code>to_numeric</code> , coerced
Decomposition	Separate trend, noise	CEEMDAN, 100 trials
IMF padding	Uniform tensor width	Zero-pad to $N_{\max} = 9$
Normalisation	Consistent scale	Min–max to $[0, 1]$
Windowing	Supervised pairs	30-step sliding window
Splitting	Prevent leakage	Chronological 80/20

scikit-learn for scaling, and Flask for the service layer.

A. Dataset and Preprocessing

The dataset comprises 1,085 multivariate observations from 1 January 2016 to 29 November 2024 at a median interval of three days, with no missing values (Table II). Turbidity is notably heavy-tailed, with a maximum of 149.8 NTU against a mean of 9.85 NTU, reflecting episodic runoff events.

The preprocessing sequence is summarised in Table III. Features and targets are scaled independently to $[0, 1]$ so predictions can be inverse-transformed for thresholding. The feature scaler’s column ordering is persisted with the model; without this, IMF columns recomputed at inference can be reordered relative to training, silently corrupting predictions.

B. Sequence Construction and Configuration

A sliding window of 30 steps predicts the observation at step 31, giving 1,055 sequences split chronologically (never randomly, which would leak future information) into 844 training and 211 test. Each input has shape 30×81 , since nine parameters each contribute nine padded IMF channels. Table IV lists the configuration; all nine parameters are output jointly from one dense layer, so a single network is trained.

At inference the service recomputes CEEMDAN over a trailing 200-observation window, reorders columns to the persisted order, and predicts from the final 30 steps. This keeps inference cost stable as the archive grows, though it introduces an inconsistency discussed in Section IV-C.

TABLE IV
MODEL ARCHITECTURE AND HYPERPARAMETERS

Component	Configuration
Input shape	30 × 81
Conv1D	64 filters, kernel 3, ReLU
MaxPooling1D	pool size 2
LSTM	50 units, ReLU, return sequences
Self-attention	tanh scoring, softmax weighting
Dense output	9 units (one per parameter)
Loss / optimiser	MSE / Adam
Epochs / batch	50 / 32

TABLE V
USER ACTIONS AND CORRESPONDING BACKEND OPERATIONS

User action	Backend	Output
Select forecast date	Prepare 30-step window	Window confirmation
Run analysis	CEEMDAN → inference	Predicted values, plots
Ask “why?”	LLM over prediction context	Natural-language summary
Export	Serialise results	CSV or PDF download

C. Interface and Chatbot Integration

The front-end is built in Streamlit, chosen for rapid iteration and native support for interactive plots. Table V traces the three principal user actions through to the backend operation each triggers. Selecting a forecast date prepares the corresponding 30-step window; running an analysis invokes CEEMDAN over the trailing window followed by model inference; and asking the chatbot for an explanation routes the current prediction, its residual context and any active alerts to the language model for interpretation.

Two design choices in this layer follow directly from the findings reported in Section IV. First, because the model was trained and validated for single-step prediction, requests beyond that horizon carry an explicit uncertainty note rather than being silently extrapolated. Second, because the analysis in Section IV-B shows the predictions to be strongly damped in amplitude, the interface presents forecasts alongside the observed historical range for the same parameter, so a user can see that a prediction sitting near the seasonal mean reflects model conservatism rather than a genuine expectation of stable conditions. Presenting a damped point forecast without that context would overstate its confidence.

IV. RESULTS AND PERFORMANCE ANALYSIS

Evaluation uses the held-out final 211 sequences, inverse-transformed to physical units before scoring. We report MAE, RMSE and R^2 against two baselines: *persistence*, the appropriate reference for a short-horizon forecaster, and *train mean*, which establishes the $R^2 = 0$ line [8].

The model reduces MAE against persistence on seven of nine parameters, with largest gains on TDS (33.5%), conduc-

TABLE VI
TEST-SET PERFORMANCE AGAINST BASELINES (211 SEQUENCES)

Parameter	MAE		RMSE		R^2
	Model	Persist.	Model	Persist.	
Temperature	3.655	0.816	4.196	1.052	0.296
pH	0.135	0.181	0.169	0.228	-0.173
Dissolved oxygen	1.093	0.331	1.362	0.455	-0.040
Conductivity	42.00	60.08	52.69	72.79	-0.050
Turbidity	6.948	8.044	13.450	18.543	-0.128
TDS	28.21	42.43	35.47	51.00	-0.011
Ammonia	0.071	0.082	0.088	0.116	-0.141
Nitrates	0.793	1.034	1.053	1.378	-0.075
Phosphates	0.040	0.050	0.050	0.070	-0.038

TABLE VII
TREND AGREEMENT AND AMPLITUDE RATIO

Parameter	Pearson r	$\sigma_{\text{pred}}/\sigma_{\text{act}}$	R^2
Temperature	0.562	0.653	0.296
Dissolved oxygen	0.481	0.727	-0.040
Conductivity	0.034	0.209	-0.050
TDS	0.066	0.190	-0.011
Turbidity	0.005	0.360	-0.128
pH	-0.010	0.276	-0.173
Ammonia	-0.031	0.166	-0.141
Nitrates	-0.061	0.074	-0.075
Phosphates	0.008	0.140	-0.038

tivity (30.1%) and nitrates (23.3%), and improves RMSE on those same seven, indicating the gain is not purely an artefact of outlier handling.

A. Trend Capture Versus Point Accuracy

The accuracy target set at the outset was not met: eight of nine parameters have $R^2 \leq 0$, so on a squared-error basis the model does not improve on predicting the training mean. Reporting this plainly matters, since much of the surveyed literature omits such baselines [5], [6].

That headline figure, however, conflates two questions: whether the model knows *when* a parameter rises and falls, and *how far*. Because R^2 penalises both jointly, a predictor with correct phase but compressed amplitude scores almost as harshly as one with no skill. We therefore report Pearson correlation r , which is scale-invariant and so measures trend agreement alone, alongside the variance ratio $\sigma_{\text{pred}}/\sigma_{\text{act}}$ in Table VII.

The outcome is sharply bimodal. For the two seasonally driven parameters the model achieves moderate positive correlation ($r = 0.562$ for temperature, 0.481 for dissolved oxygen) while reproducing 65–73% of observed variance. Temporal structure is therefore genuinely learned: Fig. 3 shows predicted temperature following the annual cycle in correct phase, and the dissolved oxygen panel of Fig. 4 tracks the mid-series depression and recovery. What the model gets wrong is magnitude, under-shooting peaks and troughs. This is the signature of a regressor trained under squared error on a partly

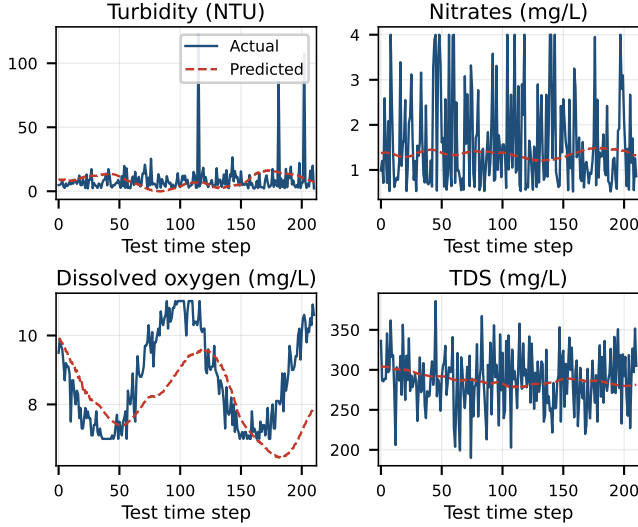


Fig. 4. Predicted (dashed) versus actual (solid) traces. Dissolved oxygen shows correct phase with damped amplitude; the remaining panels show near-constant output, reflecting parameters driven by external forcing absent from the input.

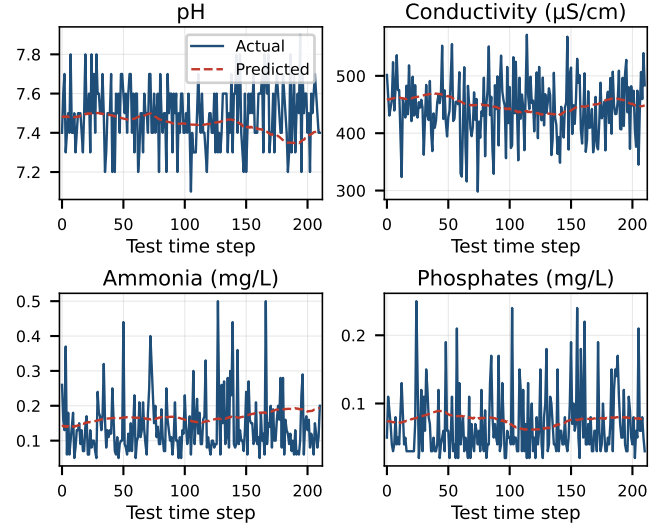


Fig. 5. Predicted (dashed) versus actual (solid) traces for the remaining four parameters. All show the near-constant output characteristic of a model predicting close to the conditional mean.

unpredictable target, where shrinking toward the conditional mean minimises loss.

For the remaining seven parameters correlation is statistically indistinguishable from zero ($|r| < 0.07$) and the variance ratio falls to 0.07–0.36, so the output is effectively constant. These are precisely the parameters governed by episodic external forcing (runoff, discharge events, sediment mobilisation) rather than the smooth seasonal cycle. Their behaviour is not encoded in their own recent history, so no purely autoregressive model, however expressive, can recover it from the inputs supplied.

Fig. 5 completes the picture for the four parameters not shown earlier. pH illustrates the central point most sharply: its observed range is narrow (7.10–7.90 across the test set, standard deviation 0.15), so a near-constant prediction achieves an MAE of only 0.135, under 2% of the mean value, while explaining none of the variance ($R^2 = -0.173$). A reader shown the error alone would judge this the model’s strongest result rather than one of its weakest.

Conductivity and TDS are physically coupled through ionic content, and in the observations they correlate at $r = 0.956$. The model’s predictions for the two correlate at only 0.679, so the network has not reproduced the strongest inter-parameter relationship in the data despite predicting both from a shared representation through a single dense layer. Capturing such coupling was part of the motivation for a joint output, and the gap suggests the shared layer is too thin to encode it.

B. Why Persistence Wins Where It Does

The split in Table VI between parameters where persistence beats the model and those where it fails badly is not arbitrary. It is predicted almost exactly by a single property of the raw

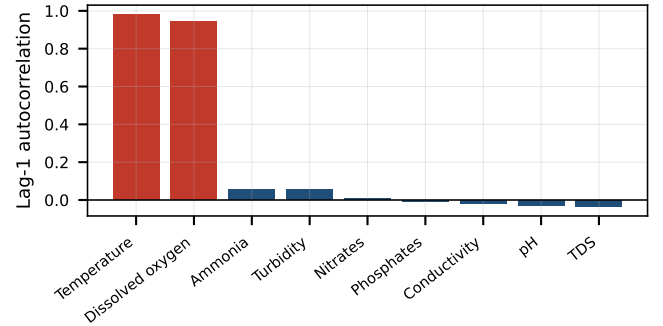


Fig. 6. Lag-1 autocorrelation of each raw series. Only temperature and dissolved oxygen (dark red) retain substantial correlation at the three-day sampling interval; the remaining seven are indistinguishable from white noise at this lag.

TABLE VIII
AUTOCORRELATION, DAMPING SLOPE AND SKILL

Parameter	Lag-1 ACF	Slope b	R^2
Temperature	0.980	0.367	0.296
Dissolved oxygen	0.943	0.350	−0.040
Turbidity	0.054	0.002	−0.128
Ammonia	0.054	−0.005	−0.141
Nitrates	0.008	−0.005	−0.075
Phosphates	−0.009	0.001	−0.038
Conductivity	−0.016	0.007	−0.050
pH	−0.031	−0.003	−0.173
TDS	−0.034	0.013	−0.011

series: the lag-1 autocorrelation, shown in Fig. 6 and tabulated in Table VIII.

Temperature and dissolved oxygen have lag-1 autocorrela-

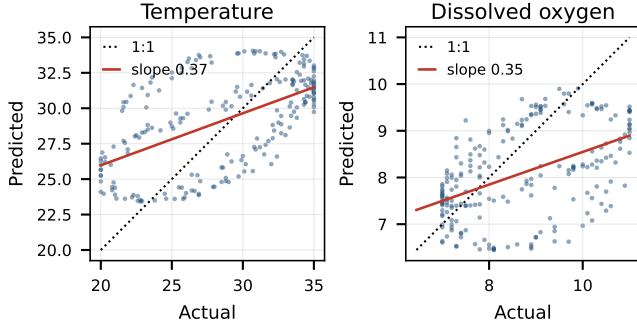


Fig. 7. Predicted against actual values for the two learnable parameters. The fitted slope (solid) falls well below the 1:1 line (dotted), quantifying the amplitude damping: the model spans roughly a third of the observed range.

tions of 0.980 and 0.943: at a three-day interval, yesterday’s reading is almost the whole answer, which is precisely why persistence attains $R^2 = 0.956$ on temperature and 0.884 on dissolved oxygen. Any model must clear a very high bar on these two, and ours does not.

The other seven parameters have $|ACF| \leq 0.054$, statistically indistinguishable from white noise at this lag. Their three-day-ahead values are effectively unpredictable from their own recent history, so persistence scores $R^2 \approx -1$ by echoing noise, and any method that shrinks toward the mean beats it. Our model’s apparent advantage on seven of nine parameters is therefore largely an artefact of the baseline being pathological on exactly those channels, not evidence of learned dynamics. This is a sobering reading of our own headline result, and we state it explicitly because the earlier comparison invites the opposite conclusion.

Column three of Table VIII quantifies the damping directly. Regressing predicted on actual values gives a slope b that would equal one for a calibrated predictor. Temperature attains $b = 0.367$ and dissolved oxygen $b = 0.350$, so both reproduce roughly a third of the observed dynamic range, visible as the flattened point cloud in Fig. 7. For the other seven, $|b| \leq 0.013$: the predictions are essentially flat, confirming that the near-zero correlations of Table VII reflect an absence of signal rather than a phase error.

Taken together, the two columns give a compact diagnostic. Lag-1 autocorrelation says whether a parameter is learnable from its own history at all; the damping slope says how much of the learnable amplitude the model actually recovered. Reporting both would let readers of the wider literature distinguish genuine dynamic modelling from mean reversion, which aggregate error metrics cannot do.

C. Diagnosis and Anomaly Detection

Two implementation defects plausibly contribute. First, **decomposition leakage**: CEEMDAN is applied to the whole series before the chronological split, so each IMF is computed from the full record and training inputs contain test-period information. Second, a **train/inference mismatch**: training decomposes the full series whereas inference uses a trailing

TABLE IX
RESIDUAL BASELINES AND 3σ ALERT THRESHOLDS

Parameter	σ	3σ	% of mean
Conductivity ($\mu\text{S/cm}$)	52.18	156.53	34.9
Turbidity (NTU)	13.43	40.30	409.3
TDS (mg/L)	35.47	106.40	36.6
Ammonia (mg/L)	0.084	0.253	169.4
Nitrates (mg/L)	1.023	3.070	207.3
Phosphates (mg/L)	0.049	0.147	213.6

Residual distributions with 3σ alert threshold (dashed)

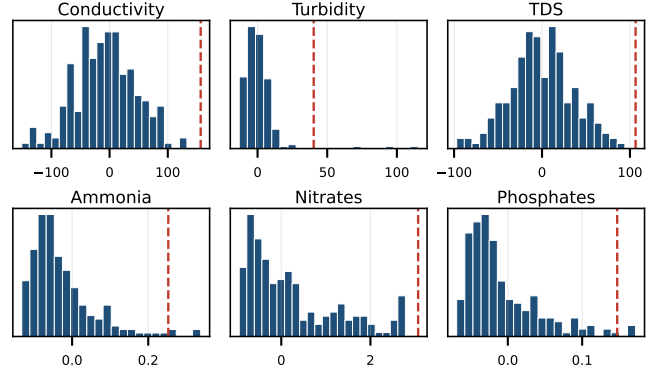


Fig. 8. Residual distributions for the six monitored parameters, with the 3σ alert threshold marked. Heavy right tails on turbidity and the nutrient species place the threshold far above the bulk of the distribution.

window, and IMFs over different support are not comparable. Zero-padding to $N_{\max} = 9$ and an unweighted MSE across jointly scaled outputs further dilute the signal.

Residual standard deviations recorded during training form the detection baseline for the six pollution-indicative parameters, listed in Table IX. Turbidity, ammonia, nitrates and phosphates are excluded from the temperature, pH and dissolved oxygen set because only they indicate contamination when elevated; a reading below prediction carries no such meaning, so the test is deliberately one-sided.

The final column of Table IX exposes a practical limitation. For conductivity and TDS the threshold sits at roughly a third of the mean value, which is a plausible alerting level. For turbidity it is over four times the mean, and for the three nutrient species between 1.7 and 2.1 times: an ammonia excursion would have to reach 0.25 mg/L above prediction before firing, against a mean of 0.149 mg/L. Fig. 8 shows why. These parameters have heavy right-tailed residual distributions, so their standard deviation is inflated by the same rare events the detector is meant to catch. The system therefore favours precision over recall, which suits an alerting context where false alarms erode operator trust, but it will miss moderate contamination. A quantile-based threshold estimated on a robust scale such as the median absolute deviation would be better matched to these distributions than a Gaussian 3σ rule.

D. Limitations

Three limitations bound the results. First, the evaluation covers a single station and a single river, so spatial generalisability is untested; the literature reviewed in Section I-B identifies this as a recurring weakness and our design does not escape it. Second, the model predicts one step ahead, and because the sampling interval is three days the effective horizon is short. Multi-horizon forecasting would require either recursive application, which compounds the amplitude damping already observed, or a direct multi-output formulation. Third, the anomaly layer has no ground truth: we can report the thresholds it derives, but without confirmed pollution incidents we cannot state a detection rate or false-alarm rate, and no claim about operational performance should be inferred from its inclusion here.

V. CONCLUSION

This work presented a CEEMDAN-CNN-LSTM framework with self-attention for multivariate river water quality prediction, coupled to a residual-based anomaly detector and deployed with automated quarterly retraining. On a 1,085-observation record the model reduced both MAE and RMSE against persistence on seven of nine parameters, by up to 33%.

The point accuracy targeted at the outset was not reached, and we report that directly. Separating trend agreement from amplitude, however, shows the headline R^2 to be misleading as a summary of what was learned. On the seasonally driven parameters the network tracks the observed trajectory in correct phase while reproducing only 65–73% of its variance: it has learned *when* these parameters move, but not *how far*. On the seven parameters governed by episodic external forcing no temporal structure is recovered, the expected outcome for an autoregressive model whose inputs contain no record of the driving events.

Both findings point the same way. Amplitude and event timing are set largely by catchment conditions such as rainfall and runoff, land use, upstream discharge and sediment transport. None of these is observable from a station's own history, but all are carried by remotely sensed catchment data [9], [10].

Beyond the architectural direction below, three implementation corrections follow directly from Section IV-C and should be made first, since each is a defect rather than a design limit: fitting CEEMDAN on the training partition alone, applying identical windowing at training and inference, and replacing the shared zero-padded tensor with per-parameter decomposition heads. A per-parameter loss weighting would additionally stop the high-variance channels from dominating the gradient.

Future work should then pair this temporal expert with a *spatial expert* rather than keep enlarging the sequence model: spectral indices derived from multispectral imagery over each catchment, reduced to a compact feature subset by evolutionary selection and regressed with a random forest suited to the small-sample, high-dimensional regime. An attention gate learning per-sample weights w_t and w_s would then arbitrate between them, deferring to the temporal stream for the seasonal parameters it already tracks and to

spatial evidence for the runoff-driven parameters where it is uninformative. Establishing that the temporal stream captures trend reliably is what makes this division of labour viable. The defects in Section IV-C should be corrected alongside it, since amplitude damping will otherwise limit the gain from fusion, and extending across multiple stations so neighbouring sites inform one another [2] remains a natural longer-term step.

REFERENCES

- [1] Q. Zhu, X. Yu, L. Zhao, and L. Xu, "A hybrid deep learning model for water quality prediction," *Journal of Water Quality Analysis*, Jun. 2025.
- [2] M. Yan and Z. Wang, "Water quality prediction method based on reinforcement learning graph neural network," *Journal of Environmental Engineering*, Dec. 2024.
- [3] P. Shu et al., "Deep learning-based super-resolution of remote sensing images for enhanced groundwater quality assessment and environmental monitoring in urban areas," *Environmental Monitoring and Remote Sensing Letters*, Mar. 2025.
- [4] Y. Sudriani, V. Sebestyén, and J. Abonyi, "Surface water monitoring systems—the importance of integrating information sources for sustainable watershed management," *Journal of Hydrology and Water Resources*, Mar. 2023.
- [5] J. K. Pandya, S. S. Khandelwal, R. K. Tipu, and K. S. Pandya, "Advancing water quality management: an integrated approach using ensemble machine learning and real-time interactive visualization," *International Journal of Water Informatics*, May 2025.
- [6] M. A. Rahu et al., "Toward design of IoT and machine learning-enabled frameworks for analysis and prediction of water quality," *International Journal of Smart Water Systems*, Sep. 2023.
- [7] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- [8] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne: OTexts, 2021.
- [9] V. Sagan et al., "Monitoring inland water quality using remote sensing: potential and limitations of spectral indices, bio-optical simulations, machine learning, and cloud computing," *Earth-Science Reviews*, vol. 205, art. 103187, 2020.
- [10] N. Pahlevan et al., "Seamless retrievals of chlorophyll-a from Sentinel-2 (MSI) and Sentinel-3 (OLCI) in inland and coastal waters," *Remote Sensing of Environment*, vol. 240, art. 111604, 2020.